

Journal of Financial Crises

Volume 4 | Issue 2

2022

Sweden: Corporate Bond Purchases

Carey K. Mott
Yale School of Management

Follow this and additional works at: <https://elischolar.library.yale.edu/journal-of-financial-crises>



Part of the [Economic Policy Commons](#), [Finance and Financial Management Commons](#), [Macroeconomics Commons](#), [Policy Design, Analysis, and Evaluation Commons](#), [Policy History, Theory, and Methods Commons](#), and the [Public Administration Commons](#)

Recommended Citation

Mott, Carey K. (2022) "Sweden: Corporate Bond Purchases," *Journal of Financial Crises*: Vol. 4 : Iss. 2, 1690-1719.

DOI: <https://doi.org/10.17132/2693-3179.1367>

Available at: <https://elischolar.library.yale.edu/journal-of-financial-crises/vol4/iss2/78>

This Case Study is brought to you for free and open access by the Journal of Financial Crises and EliScholar – A Digital Platform for Scholarly Publishing at Yale. For more information, please contact journalfinancialcrises@yale.edu.

READERS TAKE NOTICE:

As of April 2024, the YPFS Resource Library's site domain has changed to <https://elischolar.library.yale.edu/ypfs-financial-crisis-resource-library/>.

Please be aware that upon clicking any of the URL references to the former Resource Library domains, either <https://ypfs.som.yale.edu> or <https://ypfsresourcelibrary.blob.core.windows.net/>, in the "References/Key Program Documents" section of a case study, readers will encounter a "page not found" error.

Readers can still retrieve a given resource cited within this case study on the [new site](#) by searching [here](#) for the title cited.

Sweden: Corporate Bond Purchases¹

Carey K. Mott²

Yale Program on Financial Stability Case Study

July 15, 2022

Abstract

In the spring of 2020, corporate revenues in Sweden felt the direct effects of the coronavirus pandemic and the resulting public health measures. With future cash flows in question, many investors sold corporate debt for safe assets. Sweden's corporate bond market—particularly vulnerable to stress due to its heterogeneity, fragmentation, and lack of transparency—saw diminished liquidity. On March 19, 2020, the Sveriges Riksbank (Riksbank) announced it would purchase commercial paper and corporate bonds as part of a much larger bond-buying scheme, announced three days earlier, that included Swedish government, municipal, and covered bonds. It authorized the program under Chapter 6, Article 5 of the Sveriges Riksbank Act. Commercial paper purchases began soon after this announcement, but corporate bond purchases did not commence until September 2020. Although corporate credit conditions had improved by then, the Riksbank sought to establish a presence on the corporate bond market so it could scale purchases if needed. In June 2020, the Riksbank said it would limit its purchases to a nominal SEK 10 billion (about USD 1.2 billion), but it later raised the limit to SEK 13 billion. The program purchased kronor-issued, investment-grade bonds on the secondary market with maturities of less than five years. Later that year, the Riksbank announced purchases would be further limited to issuers complying with certain carbon emissions standards. Purchases peaked at SEK 445 million the week of February 15, 2021. After an extension to its original expiration date of June 30, 2021, the program formally expired on December 31, 2021. In November 2021, the Riksbank announced it would replace maturing bonds in order to maintain its net holdings, a practice it would review each quarter.

Keywords: corporate bonds, COVID-19, market liquidity, Sveriges Riksbank, Sweden

¹ This case study is part of the Yale Program on Financial Stability (YPFS) selection of New Bagehot Project modules considering market support programs in response to COVID-19. Cases are available from the *Journal of Financial Crises* at <https://elischolar.library.yale.edu/journal-of-financial-crises/>.

² Research Associate, YPFS, Yale School of Management.

Overview

Sweden, like other countries, responded to the COVID-19 pandemic with some restrictions and precautionary measures to control the health crisis (Riksbank 2020a; Riksbank 2020b; Riksbank 2020c). As corporate revenues suffered, investors worried whether many companies would survive. Equity markets became highly volatile and share prices fell dramatically (Wollert 2020; Riksbank 2020b). A flight to safe assets ensued, and liquidity on the Swedish bond market dried up. The intense selling pressure caused severe funding problems for nonfinancial corporations who funded themselves with market financing, especially those firms that needed to refinance in a stressed market (Wollert 2020).

Sweden's corporate bond market was particularly vulnerable due to the large share of unrated companies (Riksbank 2020a; Wollert 2020). The market is distinguished by a lack of transparency in pricing and trading, leading to unreliable pricing. A large share of issuance is from property companies, which were disproportionately affected by the coronavirus pandemic (Wollert 2020).

In primary markets, issuance of new bonds became expensive. For a period in March 2020, investors and issuers could not agree on prices amid a depleted investor base and no new bonds were issued (Wollert 2020). Meanwhile, risk premiums rose in secondary markets³, where the Riksbank observed a lack of liquidity and transparency amid low market activity (Riksbank 2020a).

The Riksbank sought to arrest further deterioration in credit lending by announcing dual measures to support commercial paper and corporate bonds. In a series of press releases,

Key Terms

Purpose: "... together with the purchases of commercial paper that are already in progress, [to] keep companies' funding costs down and reinforce the Riksbank's capacity to act if the credit supply to companies were to deteriorate" (Riksbank 2020q).

Launch Dates	Authorized: March 16, 2020
--------------	----------------------------

	Announced: March 19, 2020
--	---------------------------

Operational Date	September 14, 2020
------------------	--------------------

End Date	December 31, 2021
----------	-------------------

Legal Authority	Chapter 6, Article 5 of the Sveriges Riksbank Act (1988:1385)
-----------------	---

Administrators	Sveriges Riksbank
----------------	-------------------

Overall Size	Originally limited to SEK 10 billion, later raised to SEK 13 billion
--------------	--

Purchased Assets	Investment-grade corporate bonds issued in Swedish kronor by Swedish non-financial corporations
------------------	---

Peak Utilization	SEK 445 million during the week of February 15, 2021
------------------	--

³ Typically an increase in risk premia on the secondary market leads investors to increase their required rate of return for new bonds issued on the primary market, compounding problems in the Swedish bond market (Wollert 2020).

the Riksbank announced its intent to purchase up to a nominal SEK 32 billion⁴ in commercial paper beginning in April 2020 and a nominal SEK 10 billion in corporate bonds⁵ beginning in September 2020 (Riksbank 2020o; Wollert 2020). The Riksbank increased the limit on corporate bond purchases to SEK 12 billion in April 2021 and to SEK 13 billion on June 30, 2021 (Riksbank 2021e; Riksbank 2021f).

Corporate bond purchases were viewed as “a complement” to commercial paper purchases and a means for the Riksbank to establish a presence on these markets. The Riksbank said it was positioning itself to expand purchases “if monetary policy needed to be made more expansionary” (Riksbank 2020q; Riksbank, n.d.j). Purchases began on September 14, 2020. The Riksbank purchased bonds on the secondary market from its monetary policy counterparties. It limited purchases to non-subordinated corporate bonds issued by non-financial companies with credit ratings equivalent to Baa3/BBB- or higher and a remaining maturity of up to five years (Riksbank 2020q). The Riksbank only purchased kronor-issued bonds, which represent less than half of Swedish bonds (Wollert 2020; Riksbank 2020o). On December 16, 2020, the Riksbank updated its eligibility criteria, announcing it would only purchase corporate bonds issued by companies “assessed to comply with international standards and norms for sustainability” (Riksbank 2021d). In January 2021, the Riksbank began measuring the carbon footprint of its corporate bond portfolio (Riksbank 2021d; Andersson and Stenström 2021).

Riksbank’s total holdings of corporate bonds reached SEK 10 billion in late June 2021, around the originally announced expiration date of the program, and SEK 12.2 billion by October 2021, net of maturities (Riksbank, n.d.l). On November 25, 2020, the Riksbank announced it would extend its purchases of corporate bonds, as well as the other assets in its purchasing program, to December 31, 2021 (Riksbank 2020l; Riksbank 2020z).

Summary Evaluation

The market for corporate debt recovered shortly after the Riksbank announced on March 19 its intent to purchase commercial paper in April and corporate bonds sometime between March and December 2020.⁶ By June, commercial paper purchases had peaked and Riksbank’s total holdings topped SEK 6 billion. In the corporate bond market, kronor-denominated issue volumes had begun to recover and risk premiums and yields had begun to decline (Riksbank 2020f; Wollert 2020). In July, the Riksbank found that yields on safe bonds had stabilized, risky asset prices had returned to normal, and differentials between

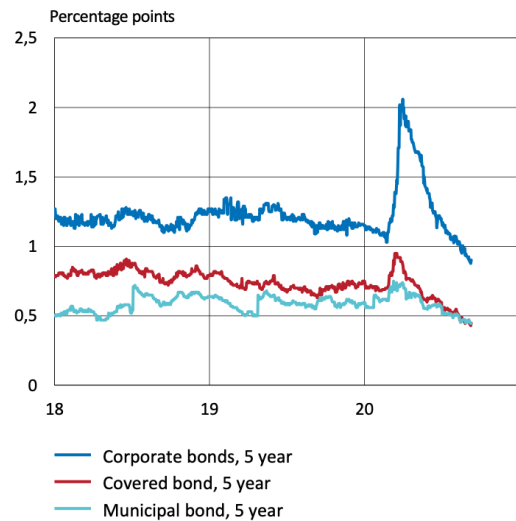
⁴ Per Yahoo Finance, USD 1 = SEK 10.01 on April 1, 2020. The Riksbank calculated its holdings of bonds in “nominal” terms; that is, based on the bond’s principal amount without including any interest payments (Riksbank, n.d.d).

⁵ In May 2020, the outstanding volumes of commercial paper and corporate bonds stood around SEK 134 billion and SEK 1.25 trillion, respectively (Riksbank 2020m, 20). For more information on the Riksbank’s commercial paper purchases, see Mott (2021).

⁶ Following the announcement in March 2020, the Riksbank conducted an analysis of the Swedish corporate bond market prior to initiating corporate bond purchases in September 2020 (Hansson and Birging 2021).

risky and safe assets had decreased (Riksbank 2020p) (see Figures 1 and 2). By August, before the Riksbank corporate bond purchases began, the Riksbank observed the Swedish market for corporate bonds was functioning in a “satisfactory” manner, which “could to some extent be due to the Riksbank’s announcement of its intention to purchase these securities” (Riksbank 2020o).

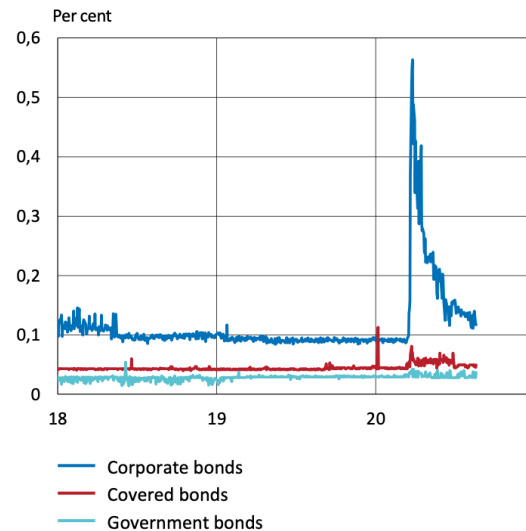
Figure 1: Yield Spread Between Bonds and Government Bonds



Data retrieved from Bloomberg, Macrobond, Refinitiv, and the Riksbank. Corporate bonds for companies with credit ratings of BBB or higher. Yield spread in relation to the government bond yield.

Source: Wollert 2020.

Figure 2: Bid-Ask Spreads for Swedish Bonds



Data retrieved from Refinitiv and the Riksbank.

Source: Wollert 2020.

Despite the improvement in market conditions, the Riksbank began its corporate bond purchases in September as planned to establish a presence in the market which would allow the Riksbank to scale up purchases in the event conditions deteriorated again, as well as to maintain its credibility by following up on its promise to commence purchases (Wollert 2022). It said that financial markets remained fragile and dependent on central bank support (Riksbank 2020m; Ingves 2021).

By the end of April, at which point the Riksbank had purchased SEK 6.3 billion in commercial paper,⁷ the central bank reported that the risk premia for investment-grade corporate bonds relative to government bonds had fallen from 2 percentage points at its peak earlier that

⁷ The Riksbank purchased commercial paper with three-month maturity (announced April 2) and six-month maturity (announced May 8). In April, outstanding commercial paper issued in Swedish kronor by non-financial corporations had an average original maturity of almost five months (Riksbank 2020m).

month to 1.6 percentage points. The Riksbank interpreted this as “an indirect indication that the [commercial paper] purchases have had an effect and that the market for commercial paper is functioning more smoothly” (Riksbank 2020h, 8). Yet a Staff Memo dated October 2020 noted that Riksbank-facilitated bank lending had largely replaced commercial paper as a source of funding for nonfinancial companies. Issuance of kronor-denominated commercial paper had not recovered to the same extent as the corporate bond market (Wollert 2020).

Figure 3: Index for Financial Conditions in Sweden

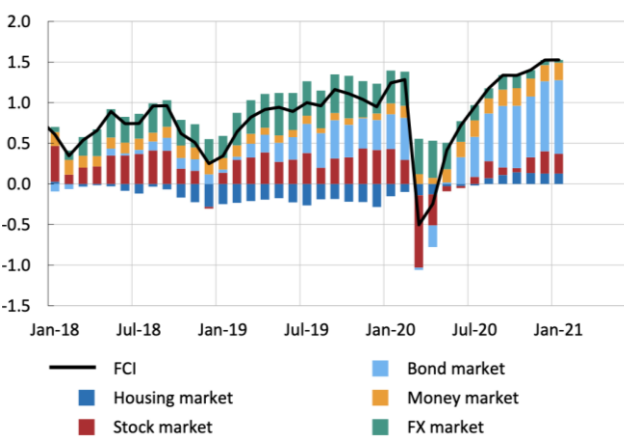
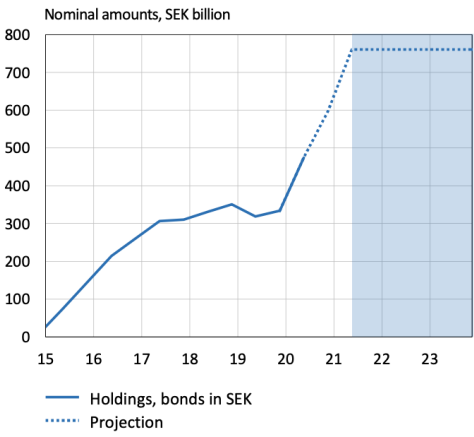


Chart displays standard deviations; a higher value indicates more expansionary financial conditions.

Source: Riksbank 2021a.

Figure 4: The Riksbank’s Holdings of Bonds



Forecast assumes entire framework will be used. Shaded area is a projection based on assumption of future decisions on purchases to compensate for maturing bond.

Source: Riksbank 2020m.

Context: Sweden 2019–2020	
GDP (SAAR, nominal GDP in LCU converted to USD)	\$534.30 billion in 2019 \$543.09 billion in 2020
GDP per capita (SAAR, nominal GDP in LCU converted to USD)	\$51,939 in 2019 \$52,274 in 2020
Sovereign credit rating (five-year senior debt)	Data for 2019: Moody's: Aaa S&P: AAAu Fitch: AAA Data for 2020: Moody's: Aaa S&P: AAAu Fitch: AAA
Size of banking system	\$744.12 billion in 2019 \$822.56 billion in 2020
Size of banking system as a percentage of GDP	139.27% in 2019 151.46% in 2020
Size of banking system as a percentage of financial system	100% in 2019 100% in 2020
Five-bank concentration of banking system	77.37% in 2019 77.09% in 2020
Foreign involvement in banking system	Data not available for 2019 Data not available for 2020
Government ownership of banking system	Data not available for 2019 Data not available for 2020
Existence of deposit insurance	Yes in 2019 Yes in 2020
<i>Sources: Bloomberg, World Bank Global Financial Development Database, and World Bank Deposit Insurance Dataset.</i>	

Key Design Decisions

1. Purpose: The Riksbank initiated corporate bond purchases to maintain the orderly function of financial markets during the COVID-19 pandemic.

The Riksbank purchased corporate bonds to “support credit supply to robust companies and prevent problems arising in financial markets that, in turn, create further problems for companies and households in need of credit.” The Riksbank also said it intended to establish a presence on the corporate bond market, “and thereby ensure a good capacity to rapidly extend the scope of the purchases if monetary policy needed to be made more expansionary” (Riksbank 2020l).

The central bank also said that buying corporate bonds would indirectly help smaller companies that didn’t have access to the bond market. Banks provide two-thirds of corporate funding in Sweden. Larger companies facing distress at the onset of the pandemic initially sought bank funding, crowding out smaller companies. If the central bank’s program would reopen the bond market for larger companies, the central bank said, smaller banks would regain access to banks (Riksbank 2020t; Gustafsson and von Brömsen 2021).

Corporate bond purchases complemented SEK 32 billion in commercial paper purchases (Riksbank 2020e; Riksbank 2020q). In Sweden, the market for corporate bonds is 10 times the size of the market for commercial paper, yet the latter is crucial for smaller companies without credit ratings (Wollert 2020). The two measures to support corporate debt markets were announced jointly at the onset of the pandemic, but corporate bond purchases did not begin until September 2020 (Riksbank 2020e; Riksbank 2020o).

2. Part of a Package: Corporate bond purchases were part of a larger asset-purchase program designed to support market functioning.

The corporate debt purchases were included in a much larger bond-purchasing program first announced on March 16. The broad program was initially set at SEK 300 billion through June 30, 2021, and was ultimately expanded to SEK 700 billion through December 31, 2021 (Riksbank 2020q; Riksbank 2021e; Riksbank 2021f). This program purchased nominal and real government bonds; bonds issued by municipalities and regions and by Kommuninvest i Sverige AB, Sweden’s local government funding agency; and covered bonds by Swedish institutions (Riksbank 2020l) (see Figure 5).

Figure 5: Riksbank Asset Purchases 2020-2021, Aggregate Holdings (SEK billions)

	2020			2021			
	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Other Support							
Government Bonds	11.5	23.5	39.5	50.7	62.8	71.0	79.8
Municipal Bonds	11.3	25.3	35.3	49.3	71.2	83.5	90.9
Covered Bonds	90.0	140.0	189.0	238.0	298.0	340.0	380.0
Treasury Bills	-	-	-	3.0	13.5	17.5	17.5
Corporate Support							
Commercial Paper	5.7	1.48	0.9	0.4	0.3	0.3	0.0
Corporate Bonds	-	0.15	1.8	4.4	7.7	10.0	11.6
Corporate lending program*	161.5	164.5	164.5	164.5	31.0	16.0	16.0

* Separate from its asset purchases, the Riksbank initiated a bank lending scheme to support corporate credit (For more information on this lending scheme, see Riksbank 2021c).

Source: Riksbank Monetary Policy Reports, 2020-2021.

On March 12, prior to announcing the corporate bond and commercial paper purchases, the Riksbank said it would offer loans to large banks against collateral totaling SEK 500 billion on particularly favorable terms if they increased lending⁸ to non-financial corporations operating in Sweden (Riksbank 2020h). Because small and medium companies in Sweden are typically customers of smaller, local banks rather than the major banks, the Riksbank also provided temporary access to the program for more counterparties than usual (Gustafsson and von Brömsen 2021).

Other state agencies provided support to businesses negatively affected by the pandemic. On March 26, Almi Företagspartner, a state venture capital program, received a capital injection of SEK 3 billion to increase lending to small and medium-sized businesses whose operations were adversely affected by the pandemic (Gustafsson and von Brömsen 2021; Regeringskansliet 2020). On April 3, the Riksgälden, the Swedish National Debt Office, issued loan guarantees for bank loans extended to affected businesses, up to SEK 75 million per borrower (Riksgälden 2020). On April 11, the European Commission approved SEK 5 billion worth of credit guarantees offered to airlines, including SEK 1.5 billion for Scandinavian Airlines, the flag carrier of Denmark, Sweden, and Norway (EC 2020a; EC 2020b). Beginning in June 2020, Skatteverket, the Swedish tax agency, granted support to companies whose revenue for March and April 2020 was less than 70% of their revenue compared to March

⁸ Bank lending has been the traditional means of funding for corporations in Sweden, as in other EU countries. But in the last decade, low funding costs for corporations and interest from investors searching for yield have combined to make wholesale funding more favorable for companies (Wollert 2020; Riksbank 2020m; Ingves 2021). This is particularly true for Sweden, whose banks have a “deposit deficit” of over SEK 3 trillion, meaning that bank deposits are insufficient to fund bank lending, with the remainder (deficit) covered by wholesale funding (Eidestedt, Forsman, and Ünlü 2020). In July 2020, wholesale funding amounted to a third of total borrowings and roughly a quarter of Swedish GDP (Riksbank 2020m).

and April 2019, and who could prove the decline was due to effects of the pandemic (Skatteverket, n.d.b; n.d.a).

To support the Riksgälden's programs, the Riksbank and Finansinspektionen, the financial supervisory authority responsible for financial regulation in Sweden, reduced the countercyclical capital buffer rate requirement from 2.5 to zero percent on March 13 (FI 2020a). According to Finansinspektionen, the reduction released capital for large and mid-size Swedish banks, corresponding to about SEK 900 billion in increased lending potential (FI 2020b; Riksbank 2020u).

3. Legal Authority: Corporate bond purchases were authorized under Chapter 6, Article 5 of the Sveriges Riksbank Act.

Chapter 6 of the Sveriges Riksbank Act contains provisions on the Riksbank's monetary policy operations. Article 5 specifies tools that the Riksbank has at its disposal for monetary policy purposes: to lend against collateral; to purchase, sell, and mediate securities, foreign exchange, and obligations linked to such assets; and to issue its own debt instruments (Andersson and Stenström 2021; Chapter 6, Article 5 of Riksbank, n.d.k). Because Riksbank purchases these securities from its monetary policy counterparties, Government Bill 1986/87:143 requires the Riksbank to ensure its monetary policy measures are competition-neutral; its bid purchases (auctions) and bilateral purchases were designed accordingly (Regeringen 1986).

The Sveriges Riksbank Act was last amended in 1999 (Riksbank 1999). In 2019, the government proposed a revised Sveriges Riksbank Act that, *inter alia*, limits the Riksbank's purchases of non-sovereign assets to extraordinary circumstances, and, according to Riksbank officials, would restrict the range of corporate bonds and other private securities that are eligible for purchase by the Riksbank (Ministry of Finance 2019; Riksbank 2020g). The proposed Sveriges Riksbank Act did not influence the Riksbank's pandemic response measures as it had not yet passed the Riksdag, the national legislature and the supreme decision-making body of Sweden; if passed, the new Act would take effect in 2023 (Riksbank, n.d.i).

In adding sustainability to its eligibility criteria,⁹ the Riksbank relied on Chapter 9 of the Sveriges Riksbank Act, which requires the Executive Board¹⁰ of the Riksbank to ensure that operations are conducted efficiently and in accordance with applicable laws, and that central government funds are managed prudently¹¹ (Andersson and Stenström 2021). The Riksbank

⁹ Since 2019, the Riksbank has performed analysis of its composition of foreign exchange reserves and evaluated how much these assets increase the amount of greenhouse gases, resulting in some assets being sold (Riksbank 2020aa).

¹⁰ The Riksbank is a public authority under the Riksdag. The Riksdag appoints the members of the Riksbank's General Council, and the General Council appoints the members of the Riksbank's Executive Board, as well as supervising the work of the Executive Board (Riksbank, n.d.k).

¹¹ The Riksbank interpreted this to mean that it must manage any risks that arise from a monetary policy measure—including credit risk—which informed the Riksbank's decision to limit purchases to investment grade, among other criteria (Andersson and Stenström 2021). In establishing that "sustainability and climate risks are now classified as sources of financial risk," the Riksbank pointed to the "Taxonomy Regulation"

used these data when purchasing corporate bonds to “avoid certain companies or sectors that are judged to be too risky from a sustainability and climate perspective,” thereby “[limiting] the Riksbank’s financial risk linked to sustainability” and fulfilling its risk management obligations (Andersson and Stenström 2021, 7–8).

4. Governance: The Executive Board of the Riksbank, the Head of the Monetary Policy Department, and the Head of the Financial Stability Department approved corporate bond purchases while the General Council informed the Riksdag of the Riksbank’s activities.

The Executive Board of the Riksbank, the Head of the Monetary Policy Department, and the Head of the Financial Stability Department approved corporate bond purchases (Riksbank 2020z).

The Riksbank is considered a public authority under the Riksdag, Sweden’s parliament, which appoints the Riksbank’s General Council (Riksdag 2021, Chapter 9, Article 13). The General Council is tasked with, among other things, enabling transparency and control for the Riksdag and keeping the Riksdag’s Committee on Finance informed of important matters at the Riksbank (Ingves and Meyersson 2021; Ministry of Finance 2019). The General Council also appoints the six-member Executive Board, including the Governor of the Riksbank, which pursuant to the Instrument of Government manages the activities of the Riksbank (Riksbank 2021, 1–2; Riksdag 2021, Chapter 9, Article 13).

The Riksbank Act states the Riksbank must deliver a monetary policy report twice a year to the Riksdag’s Committee on Finance (Riksbank, n.d.k, Chapter 6, Article 4). The Riksbank’s monetary policy reports contained details on its purchases of corporate debt (Riksbank 2020m; Riksbank 2020p; Riksbank 2020u). Prior to the Executive Board making a monetary policy decision “of major importance,” the Riksbank must inform the Finance Minister (Riksbank, n.d.k, Chapter 6, Article 3).

In 2015, the Riksdag’s Committee on Finance criticized how the Riksbank’s activities are scrutinized (Ingves and Meyersson 2021; Riksdag 2016). In 2018, the Riksdag established the Riksbank Inquiry to determine how the Riksdag could strengthen scrutiny of the Riksbank and its monetary policy (Ministry of Finance 2019). In 2019, the Riksbank Inquiry, noting the Riksbank’s high degree of independence relative to other central banks and Swedish authorities, proposed a new Sveriges Riksbank Act to the Council of Law, which if approved would take effect in 2023 (Ingves and Meyersson 2021, 5).

(2020/852) adopted by the European Council and the European Parliament in June 2020 and noted that “climate-related risks can give higher costs and lower income, that is, affect companies’ future cash flows and thereby their credit ratings” (Andersson and Stenström 2021, 7, 10). Yet, the Riksbank acknowledged that it relies on external assessments to evaluate credit risk, and “it is not certain that sustainability and climate risks are sufficiently incorporated into the rating agencies’ assessments” (Andersson and Stenström 2021, 7). So, in addition to ratings-agency assessments, the Riksbank required companies to report carbon emissions data or have these data estimated by a vendor, Sustainalytics (Riksbank 2020z; Andersson and Stenström 2021).

5. Administration: The Riksbank administered the bond purchase program and empowered the Head of the Markets Department to conduct asset purchases.

The asset purchases were a monetary policy decision made by the Executive Board (Riksbank 2020o). The Head of the Riksbank's Markets Department (with the right to delegate to another person) determined the amounts of CB purchases and set the terms and conditions for, and information on, participation in the CB purchase program (Riksbank 2020z).

Bilateral bond purchases and bid purchases (auctions) were cleared through Euroclear Sweden. Bids were made by the counterparties via the Bloomberg Bond Auction System (Riksbank 2020x).

In May 2020, given that the market situation remained uncertain, the Riksbank hired BlackRock Financial Markets Advisory to analyze the Swedish corporate bond market and assess possible design options for a potential corporate bond asset purchase program (Riksbank 2020j). The Riksbank did not make this report public.

The Head of the Markets Department was required to consult with the Heads of the Monetary Policy and Financial Stability Departments (with the right for these Heads to delegate to other persons) about the dates and amounts associated with each auction or purchase offer, and terms and conditions for, and information about the bond purchases. The Head of the Markets Department was required to consult with the Executive Board on these elements of the program at least once every other month (Riksbank 2020z). The Executive Board was the final authority on corporate bond purchases (Riksbank, n.d.).

6. Communication: The Riksbank initially intended corporate bond purchases to support credit supply to companies and restore market functioning; later, the Riksbank communicated its intent to establish a presence on the market should conditions deteriorate further.

The Riksbank intended corporate bond purchases to support credit supply to Swedish companies and restore market functioning, which it believed would restore the flow of credit to households. Once bond purchases began, the Riksbank stated its intent to establish a presence on the corporate bond market that would allow it to scale purchases if necessary and maintain expansionary monetary policy (Riksbank 2020l).

In its initial announcement, the Riksbank did not communicate a starting date for corporate bond purchases, instead indicating the purchases would begin sometime during the period of March–December 2020.

7. Disclosure: The Riksbank provided twice-yearly reports to the Riksdag that included details on the program; published aggregated auction results on a bi-weekly basis; and, beginning in 2021, published sustainability data on a monthly basis.

In announcing the details of its corporate bond purchases in August 2020, the Riksbank said it would periodically provide detailed information on its holdings of corporate bonds after it made purchases (Riksbank 2020o).

When it began its bilateral bond purchases in September 2020, the Riksbank published aggregate purchase data about those purchases on a weekly basis (Riksbank, n.d.l). Although the Riksbank published a list of companies that met its eligibility criteria, transaction-level details on individual bilateral purchases were not announced publicly in accordance with Offentlighetsoch sekretesslag (2009:400), which states that information on the Riksbank's counterparties is classified (Riksdagsförvaltningen 2009, Chapter 31, §1)

Once the Riksbank began bid purchases (auctions) on December 2, 2020, it published results on a bi-weekly basis (Riksbank, n.d.a). Directly after a completed auction, the Riksbank published information on total bid volume, number of bids, allocated volume, and price per bond (Riksbank, n.d.a; Riksdagsförvaltningen 2009 Chapter 31, §1).

On a quarterly basis beginning in January 2021, the Riksbank published the weighted average carbon intensity for its corporate bond holdings (Riksbank, n.d.b). Sustainalytics, an independent firm, calculated carbon intensity on the basis of companies' reported emissions or Sustainalytics' own estimates (Riksbank, n.d.b). The Riksbank also measured and reported on aggregate greenhouse gas emissions in the portfolio of corporate bonds, as well as emissions by industry¹² (Ferlin, Blix, and Brattström 2021).

Beginning February 1, 2021, the Riksbank published once a month, with a one-month lag, information on the volume and price per transaction for all corporate bonds the Riksbank had purchased through both the bilateral purchase procedures and by auctions (Riksbank 2020y; Riksbank, n.d.a; Riksbank, n.d.l). These publications included information on the bond issuer and the transactions details but did not name the counterparty transacting with the Riksbank (Riksbank 2020y; Riksbank, n.d.a; Riksbank, n.d.l).

Pursuant to the Sveriges Riksbank Act, the Riksbank has to submit a written report on monetary policy to the Riksdag Committee on Finance at least twice a year (Chapter 6, Article 4 of Riksbank, n.d.k). This report contained financial and operational details on the bond purchase program as well as commentary on its effects on market functioning. The Riksbank also included program details in its Annual Report submitted to the Riksdag in accordance with Chapter 10, Article 3 of the Sveriges Riksbank Act (Chapter 10, Article 3 of Riksbank,

¹² The carbon footprint for the Riksbank's holdings of corporate bonds, calculated as the weighted average carbon intensity of the portfolio, is 135 tonnes of carbon per USD million of revenue (Ferlin, Blix, and Brattström 2021).

n.d.k). The Report included an administration report, a balance sheet, a profit and loss account, and an assessment of the Riksbank's internal controls, as well as commentary on its pandemic response measures.

The Executive Board of the Riksbank usually holds five monetary policy meetings per year and publishes a Monetary Policy Report after each of these meetings (Riksbank 2021g).

8. Use of SPV: The Riksbank did not utilize a special purpose vehicle to purchase corporate bonds.

The Riksbank did not rely on a special purpose vehicle to purchase corporate bonds, instead purchasing commercial paper at auctions organized with its monetary policy counterparties.

9. Size: The Riksbank set a limit on corporate bond purchases of SEK 10 billion in 2020 but raised that limit to SEK 13 billion in 2021.

In June 2020, the Riksbank said it would limit its corporate bond purchases to SEK 10 billion (Riksbank 2020l). The Riksbank increased the limit to SEK 12 billion in April 2021 and SEK 13 billion on June 30, 2021 (Riksbank 2021e; Riksbank 2021f).

In August 2020, the Riksbank said it intended to “ensure a good capacity to rapidly adjust the scope of the purchases with consideration to their monetary policy impact and other monetary policy assessments going forward” (Riksbank 2020o, 2).

Net holdings (the total sum of bonds purchased minus maturities) topped SEK 11 billion in September 2021. The Riksbank normally held bonds it purchased to maturity (Riksbank, n.d.j; Riksbank 2022). The Riksbank used proceeds from its bond purchases to extinguish reserves.

At its peak in September 2021, the Riksbank owned 1.5% of outstanding corporate bonds on the market (Riksbank 2022). When the corporate bond purchase program began on September 14, 2020, Swedish corporations had SEK 426 billion in outstanding bonds, roughly 70% of which were investment grade and the remainder high yield (Nordic Trustee 2020l).

Bilateral corporate bond purchases peaked at SEK 455 million the week of February 15, 2021; bid purchases (auctions) peaked in June 2021, with the monthly volume offered and purchased reaching SEK 300 million and SEK 122 million, respectively (Riksbank, n.d.l).

10. Source of Funding: The Riksbank funded its corporate bond purchases through the creation of reserves.

The Riksbank purchased a bond on the secondary market and paid the seller, one of its monetary policy counterparties, by crediting the RIX account for the counterparty's bank (Kjellberg and Vestin 2019; Riksbank 2020k). The Riksbank created the money that was deposited in the banks' accounts in the RIX payment system. RIX is the Swedish payment system used by banks and other financial institutions to settle transactions. The bond

holdings were recorded as assets on the Riksbank's balance sheet. As the bond holdings matured, the reserves were extinguished (Kjellberg and Vestin 2019).

The Riksbank said that such interventions “normally have several different, and partly uncertain, consequences for the national economy and the Riksbank's own finances.” It said that the “benefits of the respective measures therefore have to be weighed against their potential drawbacks and risks” (Riksbank 2020i, 3). Although purchases of corporate debt have increased the credit risk in Riksbank's asset portfolio, the Riksbank said the risk was limited as these securities only make up a small portion of the assets. The Riksbank normally buffers financial risk with equity and unrealized gains in revaluation accounts, but the total buffer did not grow at the same rate as the Riksbank's financial risks increased. As a result, the Executive Board of the Riksbank made a provision for financial risks, allowing the Riksbank to retain SEK 5 billion in profit on its balance sheet by reducing the dividends paid to the government (Riksbank 2020a).

11. Eligible Institutions: Monetary Policy Counterparties who completed Registration were eligible to make a bid in the Riksbank's bond purchase program.

The Riksbank conducts monetary policy with the assistance of 24 monetary policy counterparties¹³, which are credit institutions either domiciled or with a branch in Sweden and that participate in Sweden's payment system, RIX (Riksbank, n.d.h; Riksbank, n.d.f). Monetary policy counterparties were required to register for each individual monetary policy measure before they could participate. Registration bound the counterparty to the terms and conditions of each measure and included contact details, Euroclear account information, and, for bid purchases, contact information for the counterparty's account in the Bloomberg Bond Auction System (Riksbank 2020x).

12. Auction or Standing Facility: The Riksbank initially purchased bonds over the counter and later organized reverse auctions.

When designing its corporate bond purchase program, the Riksbank considered the “special circumstances” of Sweden's market: its small size, fragmentation, heterogeneity, limited liquidity, and lack of transparency—all of which made it difficult for Riksbank to evaluate corporate bonds (Riksbank 2020o). These factors informed the Executive Board's decision to employ both over-the-counter purchases (which the Riksbank called “bilateral”) and bid (auction) purchases. The Riksbank pursued the two purchase methods to maintain flexibility in its choice of bonds (Riksbank 2020p; Riksbank 2020o). Both methods were available only to monetary policy counterparties who registered for the program with the Riksbank (Riksbank 2020x; Riksbank 2020v).

¹³ Like corporate bond markets in the U.S., EU, and elsewhere, there are no formal market makers in the Swedish corporate debt market like there are in the markets for government and covered bonds. However, unlike those markets, the Swedish market has relatively less liquidity due to fewer, smaller participants and lesser repo and assurance possibilities (Wollert 2020).

The Riksbank announced that its first corporate bond purchases in September 2020 would be made through over-the-counter purchases¹⁴ (Riksbank 2020r; Section D.3 of Riksbank 2020n). Although most of the Riksbank's non-corporate asset purchases were conducted through auctions, over-the-counter purchases are the most common transaction method in Sweden's corporate debt market. The Riksbank used this method first in order "to establish the Riksbank's presence on the market and thereby a good capacity to influence pricing on this market" (Hansson and Birging 2021; Riksbank 2020p). Delivery and payment were expected two days after the purchase date (Riksbank 2020r).

In its announcement of the bond purchase program in September, the Riksbank said it may purchase corporate bonds by a bid (auction) procedure "at a later stage" (Riksbank 2020q). Bid purchases were announced on November 17, 2020, and began on December 2, 2020 (Riksbank 2020y). Bid purchases were determined to be the "most market-neutral way"¹⁵ of ensuring purchases had a broad impact on interest rates (Riksbank 2020z, 4).

Before each auction, the Riksbank published the specific securities it was offering to buy and the purchase amounts. The Head of the Markets Department determined these lists after consultation with the Heads of the Monetary Policy and Financial Stability Departments (Hansson and Birging 2021). Bids submitted by monetary policy counterparties included information about the desired volume and the interest rate bid for each bond (Riksbank 2020w). When the auction concluded, the bids were ranked according to the offered interest rate. The Riksbank then decided on the final volume¹⁶ and bonds were allocated using the unit price method, whereby all allocated bids received the lowest successful rate¹⁷ (Riksbank 2020w). The counterparty that offered to sell at the highest interest rate (i.e., the lowest price) received its allocation first (Hansson and Birging 2021).

13. Loan or Purchase: The Riksbank purchased investment-grade corporate bonds on the secondary market.

The Riksbank purchased investment grade corporate bonds on the secondary market.

¹⁴ Bilateral transactions are made outside regulated marketplaces; these are also known as "over-the-counter" or OTC-traded. Most of the transactions on the secondary market are made bilaterally between buyer and seller over the telephone or an online chat (Wollert 2020). In a typical bilateral purchase, either the counterparty can be contacted directly by the Riksbank, or the transaction is executed through a market agent (Riksbank 2020n).

¹⁵ To evaluate market neutrality in the Riksbank's purchases, holdings of corporate bonds were regularly compared with a portfolio consisting of all the bonds that the Riksbank had been able to purchase since September 2020 and which had not yet matured (Riksbank, n.d.j).

¹⁶ Volumes were stated as a nominal SEK 1 million, or SEK 2 million in bonds where the smallest denomination is SEK 2 million (Riksbank 2020w). Final volumes fell within the highest and lowest permitted bid volume for each bond (Riksbank 2020w).

¹⁷ The bids with the highest submitted interest rate/credit margin mark-up were allocated first, then the next highest bids for rate/credit margin mark-up, and so on until the final volume was filled (Riksbank 2020w). The lowest accepted rate/credit margin mark-up applied to all bids allocated (Riksbank 2020w).

14. Eligible Assets: The Riksbank purchased non-subordinated corporate bonds issued in Swedish kronor by non-financial companies with investment-grade credit ratings and a remaining maturity of up to five years.

The Riksbank purchased corporate bonds from non-financial corporations for the first time in response to the COVID-19 pandemic.¹⁸ In its definition of “non-financial” corporations, the Riksbank included captive financial institutions and investment companies with a broad ownership diversification (Riksbank, n.d.g). Firms with primarily financial operations, or under the supervision of Finansinspektionen, were excluded from participating in the program (Riksbank, n.d.g).

The Riksbank purchased corporate bonds from investment-grade companies, defined as those with credit ratings of at least BBB-/Baa3 from Standard & Poor's, Moody's, Fitch Ratings, Nordic Credit Rating, or Scope Ratings (see Figure 6). The credit rating of the security was considered first, and if the bond was unrated then the company's credit rating was used (Riksbank 2020o).

Figure 6: Riksbank Corporate Bond Portfolio Ratings Composition

Credit Rating ^A	Amount (SEK million)	Percent of Riksbank's holdings	Average remaining time to maturity (months)
AA ^A	693	7%	23.9
A ^B	4,556	43%	22.9
BBB ^C	5,445	51%	24.4
High Yield	-	-	-
Not rated	-	-	-

^A Ratings by S&P, Fitch, Moody's, Nordic Credit Rating, and Scope.

^B AA+/AA/AA-

^C A+/A/A-, A1/A2/A3

^D BBB+/BBB/BBB-, Baa1/Baa2/Baa3

Data retrieved from Riksbank, Bloomberg as of Aug. 2, 2021.

Source: Riksbank, n.d.e; Riksbank, n.d.j; Wollert 2020.

The Riksbank sought a portfolio of corporate bonds that was representative of the outstanding volume eligible for purchase. As of September 2021, 58 companies were eligible under the Riksbank's criteria (Riksbank, n.d.g). In the event a security or company no longer

¹⁸ The Riksbank considered purchasing corporate bonds, among other asset classes, in its response to the Global Financial Crisis (Riksbank 2020d).

met the criteria, the Riksbank immediately excluded further purchases of these bonds, but it did not normally sell bonds (Riksbank, n.d.g).

On the date of purchase, eligible bonds must have been issued at least 30 days prior to the purchase date¹⁹, and on that date could not have a time to maturity of less than six months or more than five years (Riksbank, n.d.g).

Most corporate bonds in Sweden are issued senior unsecured. Corporate bonds are issued by Swedish companies in kronor (SEK 553 billion) and other currencies, including euros, US dollars, Norwegian krone, and pounds sterling (totaling SEK 655 billion); the Riksbank only purchased bonds issued in kronor²⁰ (Wollert 2020).

In January 2021, after “developing methods to identify risks linked to sustainability that can affect the conditions for the Riksbank’s execution of its tasks,” the Riksbank began requiring companies to comply with international standards and norms for sustainability in order to be eligible for bond purchases (Riksbank 2021d). The Riksbank employed Sustainalytics, a provider of sustainability analysis data, to perform a norms-based negative screening of the companies whose bonds were included in the Riksbank’s portfolio (Riksbank 2021d). Once the carbon footprint of each company was measured, bond purchases were limited to those companies found to be compliant. If the Riksbank had previously purchased bonds from a company found to be noncompliant, the Riksbank could sell the bonds and these bonds would be ineligible for future purchases (Andersson and Stenström 2021). (See Key Design Decision 21 for more details on sustainability criteria.)

15. Purchase Price: Bilateral purchase prices were negotiated with the seller; bid purchase prices were determined in competitive auction; no haircuts were applied.

In the bilateral purchase procedure, the relevant securities, volumes, and prices were negotiated for each transaction between the Riksbank and the monetary policy counterparty. In these negotiations, the Riksbank would reference offered prices and dealer inventories to determine a fair price for a given bond. In the bid purchase procedure, the auction involved monetary policy counterparties competing against one another in bidding with regard to the price and volume at which they were prepared to sell a particular bond (Riksbank n.d.c). (See Key Design Decision 12 for more detail on bid purchase procedures.)

16. Haircuts: This Key Design Decision is not applicable to asset purchase programs.

This Key Design Decision is not applicable to asset purchase programs.

¹⁹ To ensure bonds were not obtained from the issuer “solely to fulfill the delivery to the Riksbank,” the Riksbank stipulated that “delivery may not consist of bonds where the counterparty has purchased the bond from the issuer later than thirty days prior the purchase date” (Riksbank 2020r).

²⁰ Ownership of corporate bonds issued in Swedish kronor is spread across mutual funds (30 percent), insurance and pension funds (20 per cent), and foreign investors (30 percent). Foreign investors hold the overwhelming majority of Swedish corporate bonds issued in foreign currencies (Riksbank 2020m).

17. Interest Rates: The Riksbank did not charge an interest rate for purchases of corporate bonds.

As an asset purchase program, there was no interest rate associated with the Riksbank's corporate bond purchases.

18. Fees: The Riksbank charged normal RIX participant fees.

All RIX participants paid fees to the Riksbank according to a price list published by the Riksbank (Riksbank 2020n; Riksbank 2021b). RIX is Sweden's real-time gross settlement payments system. Fees were based on the principles of limitation of systemic risks, cost coverage, and non-discrimination (Riksbank 2020n).

Additionally, monetary policy counterparties were expected to hold a sufficient amount of securities or currency to carry out a transaction. Any counterparties unable to deliver or pay for their securities were charged interest on the part of the delivery or payment that had not been executed, at a rate equal to the current reference rate as defined in Section 9 of the Interest Act (9 § Räntelagen) plus 8 percentage points. This interest would be charged for the term equal to the intended transaction (Riksbank 2020n).

19. Term: This Key Design Decision does not apply to asset purchase programs.

This Key Design Decision does not apply to asset purchase programs.

20. Other Restrictions on Eligible Participants: The Riksbank held no more than 50 percent of an individual issuer's total outstanding bonds, and no more than 50 percent of an individual bond issuance.

The Riksbank limited its holdings of corporate bonds to no more than 50 percent of an individual company's total outstanding volume of corporate bonds in Swedish kronor and no more than 50 percent of an individual bond issuance²¹. These limits were applied to ensure the bonds purchased "proportionately reflect the total volume of securities covered," and thus affect the market in "a broad and market-neutral manner and thereby having the desired monetary policy impact" (Riksbank 2020o).

21. Other Restrictions on Eligible Participants: Beginning in January 2021, the Riksbank applied a norms-based negative screening to measure the carbon footprint of companies otherwise eligible for its bond purchases.

On December 16, 2020, the Riksbank announced it would measure and report the carbon footprint of its corporate bond portfolio and, beginning in January 2021, only purchase assets from companies that complied with international standards and norms for

²¹ For commercial paper purchases, the limit was 70 percent per issuer.

sustainability²² (Riksbank 2020; Riksbank 2020z). The negative screening was intended to help Riksbank “manage government funds prudently” by “[reducing] the financial risks that could follow from climate change or from companies failing to comply with international standards and norms” as well as “safeguarding long-term sustainable economic development” (Riksbank 2020aa, 4). The Riksbank was also motivated by adhering to “the state’s core values” and reputational concerns (Andersson and Stenström 2021, 8).

The Riksbank employed negative screening, or the exclusion of companies on the basis of certain criteria. The basis of the Riksbank’s negative screening were the principles included in the UN Global Compact, the OECD Guidelines for Multinational Enterprises, and the UN Guiding Principles on Business and Human Rights, which together included standards in the areas of human rights, labor conditions, environment, and anti-corruption (Andersson and Stenström 2021).

Recognizing that there is no standardized method for measuring sustainability and climate-related risks, and that information on companies’ climate-related risks was “insufficient, incomplete and inconsistent,” the Riksbank sought to assist companies lacking the capacity to self-report by employing a vendor, Sustainalytics, to estimate these data for companies²³ (Andersson and Stenström 2021; Riksbank 2021d). Once the external assessment was completed, the Riksbank would “make its own assessments based on the analyses produced” (Andersson and Stenström 2021, 9).

In selecting bonds according to their sustainability assessment, the Riksbank had the “aim of avoiding distorting effects on the Swedish market for these securities” (Riksbank, n.d.g). If a company whose bonds the Riksbank had already purchased were to be assessed as in breach of the norms in the future, the Riksbank would “take a stance on how to handle this taking into account the customary monetary policy considerations,” potentially “refraining from making further purchases, or selling the bonds purchased earlier” (Andersson and Stenström 2021, 9).

In addition to adding sustainability to its eligibility criteria, the Riksbank also reported the aggregate carbon footprint of its bond holdings (Riksbank, n.d.b; Ferlin, Blix, and Brattström 2021).

22. Regulatory Relief: The Riksbank did not offer its monetary policy counterparties regulatory relief for participation in its corporate bond purchases.

The Riksbank did not offer regulatory relief to participants in its corporate bond purchase program.

²² The Riksbank also announced purchases of Swedish sovereign and municipal “green bonds;” however, regarding its corporate bonds, the Riksbank placed “no particular importance on whether or not the assets are classified as green” (Riksbank 2020aa, 5).

²³ Only 40 percent of Swedish companies otherwise eligible for corporate bond purchases reported greenhouse gas emissions, meaning more than half of the portfolio’s total carbon footprint was estimated (Ferlin, Blix, and Brattström 2021, 18).

23. International Coordination: The Riksbank did not coordinate with other countries for its corporate bond purchases.

While central banks in other countries purchased corporate bonds over the same period, the Riksbank's program was not coordinated with these efforts, and its approach was specifically tailored to the distinguishing features of Sweden's corporate bond market.

24. Duration: Initially scheduled to end December 2020, the corporate bond purchase program was extended to December 31, 2021.

The Riksbank originally announced bond purchases up to SEK 300 billion through December 2020 (Riksbank 2020e). On June 30, 2020, bond buying was extended to June 30, 2021, and the limit increased to SEK 500 billion including the SEK 300 billion previously announced. In this June 30, 2020, announcement, the Riksbank communicated its intent to purchase up to SEK 10 billion in non-financial corporate bonds between September 1, 2020, and June 30, 2021 (Riksbank 2020l). On August 31, 2020, the Riksbank confirmed that bilateral corporate bond purchases would begin on September 14, 2020 (Riksbank 2020o). Then, on November 25, 2020, the Riksbank extended the bond buying program, including the purchase of corporate bonds, to December 31, 2021, and up to a total amount of SEK 700 billion, inclusive of prior purchases (Riksbank 2020l). In April 2021 and June 2021, the limits on corporate bond purchases were raised by a SEK 2 billion and SEK 1 billion, respectively, capping the program at SEK 13 billion until its expiration on December 31, 2021 (Riksbank 2020q; Riksbank 2021e; Riksbank 2021f). In November 2021, the Riksbank announced it would replace maturing bonds in order to maintain its net holdings, a practice it would review each quarter (Riksbank 2021h).

References and Key Program Documents

Program Summaries

(Riksbank 2020e) Sveriges Riksbank (Riksbank). March 19, 2020. "Purchases of Corporate Bonds and Commercial Paper for Monetary Policy Purposes."

Press release on the Riksbank's purchase of corporate debt.

<https://ypfs.som.yale.edu/library/document/purchases-corporate-bonds-and-commercial-paper-monetary-policy-purposes>

(Riksbank 2020o) Sveriges Riksbank (Riksbank). August 31, 2020. "Corporate Bond Purchases as a Monetary Policy Measure."

Announcement of purchases of corporate bonds in the week beginning September 14, 2020.

<https://ypfs.som.yale.edu/library/document/corporate-bond-purchases-monetary-policy-measure>

(Riksbank 2020z) Sveriges Riksbank (Riksbank). November 25, 2020. "Programme for the Riksbank's Asset Purchases for Monetary Policy Purposes in 2021."

Summary of the Riksbank's intended asset purchases in response to the COVID-19 pandemic.

<https://ypfs.som.yale.edu/library/document/programme-riksbanks-asset-purchases-monetary-policy-purposes-2021>

Implementation Documents

(Andersson and Stenström 2021) Andersson, Magnus, and Mikael Stenström. March 15, 2021. "Sustainability Considerations When Purchasing Corporate Bonds." *Economic Commentary*.

Riksbank staff provide an overview of the incorporation of sustainability factors in corporate bond purchases.

<https://ypfs.som.yale.edu/library/document/sustainability-considerations-when-purchasing-corporate-bonds>

(Riksbank 2020) Sveriges Riksbank (Riksbank). December 17, 2020. "Asset Management and Sustainability."

Overview of the Riksbank's approach to including sustainability as a factor in its asset purchases.

<https://ypfs.som.yale.edu/library/document/asset-management-and-sustainability>

(Riksbank 2020f) Sveriges Riksbank (Riksbank). March 26, 2020. "Press Release: Purchases of Commercial Paper."

Announcement of commercial paper purchases.

<https://ypfs.som.yale.edu/library/document/announcement-commercial-paper-purchases>

(Riksbank 2020i) Sveriges Riksbank (Riksbank). May 8, 2020. "Continuing Purchases of Commercial Paper."

Announcement of continuing commercial paper purchases.

<https://ypfs.som.yale.edu/library/document/annex-minutes-commercial-paper-purchases-continue>

(Riksbank 2020j) Sveriges Riksbank (Riksbank). May 15, 2020. "Riksbank Uses External Consultant for Analysis Work."

Press release announcing the use of BlackRock's Financial Markets Advisory for corporate bond market analysis.

<https://ypfs.som.yale.edu/library/document/riksbank-uses-external-consultant-analysis-work>

(Riksbank 2020l) Sveriges Riksbank (Riksbank). June 30, 2020. "Extended Programme for the Riksbank's Purchases of Bonds for Monetary Policy Purposes."

Announcement of bond purchase program's extension to June 30, 2021, up to 500 billion SEK.

<https://ypfs.som.yale.edu/library/document/extended-programme-riksbanks-purchases-bonds-monetary-policy-purposes>

(Riksbank 2020n) Sveriges Riksbank (Riksbank). July 2020. "Terms and Conditions for RIX and Monetary Policy Instruments."

Terms and conditions for corporate bonds purchased through bilateral purchase and bid procedure.

<https://ypfs.som.yale.edu/library/document/terms-and-conditions-rix-and-monetary-policy-instruments>

(Riksbank 2020q) Sveriges Riksbank (Riksbank). September 1, 2020. "Riksbank to Begin Purchasing Corporate Bonds in September."

Announcement initiating corporate bond purchases.

<https://ypfs.som.yale.edu/library/document/riksbank-begin-purchasing-corporate-bonds-september>

(Riksbank 2020r) Sveriges Riksbank (Riksbank). September 9, 2020. "General Terms and Conditions for the Riksbank's Purchases of Corporate Bonds via Bilateral Purchase Procedure."

General terms and conditions for the Riksbank's over-the-counter corporate bond purchases.

<https://ypfs.som.yale.edu/library/document/general-terms-and-conditions-riksbanks-purchases-corporate-bonds-bilateral>

(Riksbank 2020t) Sveriges Riksbank (Riksbank). November 2020. "Minutes of the Monetary Policy Meeting Held on 25 November 2020."

Announcements from the Monetary Policy Committee following the November 2020 policy meeting.

<https://ypfs.som.yale.edu/library/document/minutes-monetary-policy-meeting-held-25-november-2020>

(Riksbank 2020v) Sveriges Riksbank (Riksbank). November 17, 2020. "FAQs on Registration."

FAQs on registration to be a monetary policy counterparty.

<https://ypfs.som.yale.edu/library/document/questions-and-answers-about-registration>

(Riksbank 2020w) Sveriges Riksbank (Riksbank). November 17, 2020. "General Terms and Conditions for the Riksbank's Purchases of Corporate Bonds via Bid Procedure."

General terms and conditions for the Riksbank's corporate bond auctions.

<https://ypfs.som.yale.edu/library/document/general-terms-and-conditions-riksbanks-purchases-corporate-bonds-bid-procedure>

(Riksbank 2020x) Sveriges Riksbank (Riksbank). November 17, 2020. "Registration: Participation in Bid Procedures for the Riksbank's Purchase of Corporate Bonds."

Bidder registration for corporate bond purchases via auction.

<https://ypfs.som.yale.edu/library/document/registration-riksbanks-purchase-corporate-bonds>

(Riksbank 2020y) Sveriges Riksbank (Riksbank). November 17, 2020. "Riksbank Initiates Corporate Bond Purchases via Auctions and Increases Transparency."

The Riksbank's announcement that it will organize auctions for its corporate bond purchases.

<https://ypfs.som.yale.edu/library/document/riksbank-initiates-corporate-bond-purchases-auctions-and-increases-transparency>

(Riksbank 2020aa) Sveriges Riksbank (Riksbank). December 16, 2020. "Sustainability Strategy for the Riksbank."

Details on the Riksbank's approach to sustainability in its operations and policy.

<https://ypfs.som.yale.edu/library/document/sustainability-strategy-riksbank>

(Riksbank 2021b) Sveriges Riksbank (Riksbank). March 2021. "Charges for the RIX System as of 1 March 2021."

Fees charged for use of the RIX system.

<https://ypfs.som.yale.edu/library/document/charges-rix-system-1-march-2021>

(Riksbank 2021c) Sveriges Riksbank (Riksbank). March 10, 2021. "The Riksbank Is Starting a New Programme for Funding to Banks to Support Corporate Lending."

Press release detailing the Riksbank's corporate lending scheme, separate from its asset purchases.

<https://ypfs.som.yale.edu/library/document/riksbank-starting-new-programme-funding-banks-support-corporate-lending>

(Riksbank 2021d) Sveriges Riksbank (Riksbank). March 15, 2021. "Riksbank Takes Sustainability into Account When Purchasing Corporate Bonds."

Announcement that the Riksbank will only offer to purchase bonds issued by companies deemed to comply with international standards and norms for sustainability, effective January 2021.

<https://ypfs.som.yale.edu/library/document/riksbank-takes-sustainability-account-when-purchasing-corporate-bonds>

(Riksbank, n.d.a) Sveriges Riksbank (Riksbank). No date. "Auction results (Corporate bonds)." <https://ypfs.som.yale.edu/library/document/auction-results-corporate-bonds>. Accessed October 13, 2021.

Results of the Riksbank's corporate bond purchases via auction, updated bi-monthly.
<https://ypfs.som.yale.edu/library/document/auction-results-corporate-bonds>

(Riksbank, n.d.b) Sveriges Riksbank (Riksbank). No date. "Carbon Footprint of the Holdings of Corporate Bonds." Accessed September 21, 2021.

The carbon footprint of the Riksbank's corporate bond portfolio.

<https://ypfs.som.yale.edu/library/document/carbon-footprint-holdings-corporate-bonds>

(Riksbank, n.d.c) Sveriges Riksbank (Riksbank). No date. "FAQ about Extended Purchases of Securities." Accessed September 20, 2021.

FAQ document addressing how the Riksbank will handle its asset purchases and holdings in 2022.

<https://ypfs.som.yale.edu/library/document/faq-about-extended-purchases-securities>

(Riksbank, n.d.d) Sveriges Riksbank (Riksbank). No date. "Glossary." Accessed January 18, 2022.

The Riksbank's general glossary, including its definition of "nominal" accounting.

<https://ypfs.som.yale.edu/library/document/sveriges-riksbank-glossary>

(Riksbank, n.d.e) Sveriges Riksbank (Riksbank). No date. "Holdings of Corporate Bonds – per Credit Rating." Accessed September 21, 2021.

The Riksbank's holdings of corporate bonds by credit rating.

<https://ypfs.som.yale.edu/library/document/holdings-corporate-bonds-credit-rating>

(Riksbank, n.d.f) Sveriges Riksbank (Riksbank). No date. "How to Become a Monetary Policy Counterparty." Accessed November 10, 2021.

Details on becoming a counterparty to the Riksbank's monetary policy operations.

<https://ypfs.som.yale.edu/library/document/how-become-monetary-policy-counterparty>

(Riksbank, n.d.g) Sveriges Riksbank (Riksbank). No date. "List of Companies with Bonds That Meet the Riksbank's Criteria." Accessed September 20, 2021.

Identified which companies with outstanding bonds meet the Riksbank's formal criteria, updated monthly.

<https://ypfs.som.yale.edu/library/document/list-companies-bonds-meet-riksbanks-criteria>

(Riksbank, n.d.h) Sveriges Riksbank (Riksbank). No date. "Monetary Policy Counterparties." Accessed September 21, 2021.

List of the Riksbank's monetary policy counterparties.

<https://ypfs.som.yale.edu/library/document/monetary-policy-counterparties>

(Riksbank, n.d.i) Sveriges Riksbank (Riksbank). No date. "Purchases of Corporate Bonds." Accessed September 21, 2021.

Announcement of the Riksbank's corporate bond purchases.

<https://ypfs.som.yale.edu/library/document/purchases-corporate-bonds>

(Riksbank, n.d.l) Sveriges Riksbank (Riksbank). No date. "Weekly Purchases (Corporate Bonds)." Accessed September 23, 2021.

Results of the Riksbank's corporate bond purchases via bilateral procedures, updated weekly.
<https://ypfs.som.yale.edu/library/document/weekly-purchases-corporate-bonds>

Legal/Regulatory Guidance

(Kjellberg and Vestin 2019) Kjellberg, David, and David Vestin. 2019. "The Riksbank's Balance Sheet and Financial Independence." Sveriges Riksbank Economic Review.

Riksbank staff paper explaining the Riksbank's use of its balance sheet.

<https://ypfs.som.yale.edu/library/document/riksbanks-balance-sheet-and-financial-independence>

(Ministry of Finance 2019) Ministry of Finance. November 29, 2019. "The Riksbank Committee's Proposal for a New Riksbank Act, Summary."

Official report from the committee convened to consider a new Riksbank Act.

<https://ypfs.som.yale.edu/library/document/riksbank-committees-proposal-new-riksbank-act>

(Regeringen 1986) Regeringen. 1986. "Government Bill 1986/87: 143 on a New Riksbank Act and Changed Leadership for the Debt Office."

Government bill establishing the Riksbank Act.

<https://ypfs.som.yale.edu/library/document/government-bill-198687-143-new-riksbank-act-and-changed-leadership-debt-office>

(Regeringskansliet 2020) Regeringskansliet, Regeringen och. March 26, 2020. "Almi Receives SEK 3 Billion Capital Injection to Increase Lending to Small and Medium-Sized Enterprises." Text. Regeringskansliet. Regeringen och Regeringskansliet.

Details on Almi Företagspartner AB's capital injection to support SME lending.

<https://ypfs.som.yale.edu/library/document/almi-receives-sek-3-billion-capital-injection-increase-lending-small-and-medium>

(Riksbank 1999) Sveriges Riksbank (Riksbank). 1999. "New Law Gives the Riksbank More Independence."

The Riksbank's reaction to the update Riksbank Act in 1999.

<https://ypfs.som.yale.edu/library/document/new-law-gives-riksbank-more-independence>

(Riksbank 2020g) Sveriges Riksbank (Riksbank). April 28, 2020. "The Proposed Act Would Restrict the Riksbank's Independence."

The Riksbank's reaction to the proposed new Riksbank Act.

<https://ypfs.som.yale.edu/library/document/proposed-act-would-restrict-riksbanks-independence>

(Riksbank 2021) Sveriges Riksbank (Riksbank). January 15, 2021. "Rules of Procedure for Sveriges Riksbank."

The General Council's rules for the organization and management of the Riksbank.

<https://ypfs.som.yale.edu/library/document/rules-procedure-and-instructions-sveriges-riksbank>

(Riksbank, n.d.) Sveriges Riksbank (Riksbank). No date. "Instructions for Sveriges Riksbank." Webpage.
The General Council's detailed provisions for the Riksbank's organization and management.
<https://ypfs.som.yale.edu/library/document/instructions-sveriges-riksbank>

(Riksbank, n.d.i) Sveriges Riksbank (Riksbank). No date. "Proposal for a New Sveriges Riksbank Act." Accessed October 14, 2021.
Summary of the proposed changes to the Sveriges Riksbank Act.
<https://ypfs.som.yale.edu/library/document/proposal-new-sveriges-riksbank-act>

(Riksbank, n.d.k) Sveriges Riksbank (Riksbank). No date. The Sveriges Riksbank Act. Accessed September 20, 2021.
The complete Sveriges Riksbank Act, as revised in 1999.
<https://ypfs.som.yale.edu/library/document/sveriges-riksbank-act>

(Riksdag 2016) Riksdag. January 19, 2016. "Review of the Riksbank's Monetary Policy 2010–2015."
Riksdag review of the Riksbank's activities, out of which came the proposal for a new Sveriges Riksbank Act.
<https://ypfs.som.yale.edu/library/document/review-riksbanks-monetary-policy-2010-2015>

(Riksdag 2021) Riksdag. January 26, 2021. "The Instrument of Government."
Information detailing the Riksdag's oversight of the Riksbank's activities.
<https://ypfs.som.yale.edu/library/document/instrument-government>

(Riksdagsförvaltningen 2009) Riksdagsförvaltningen. 2009. Offentlighetsoch Sekretesslag (2009: 400).
Public Access and Secrecy Act of 2009, outlining confidentiality of counterparties.
<https://ypfs.som.yale.edu/library/document/offentlighetsoch-sekretesslag-2009400>

Press Releases/Announcements

(EC 2020a) European Commission (EC). April 11, 2020. "State Aid: Coronavirus: €455 Million Swedish Guarantee Scheme." European Commission - European Commission.
European Commission approval of a Swedish guarantee scheme to support airlines affected by the COVID-19 pandemic.
<https://ypfs.som.yale.edu/library/document/state-aid-coronavirus-eu455-million-swedish-guarantee-scheme>

(EC 2020b) European Commission (EC). April 24, 2020. "State Aid: Commission Approves Swedish Public Guarantee."
European Commission approval of Swedish public guarantee of up to €137 million to compensate airlines for damage caused by the COVID-19 pandemic.

<https://ypfs.som.yale.edu/library/document/state-aid-commission-approves-swedish-public-guarantee-eu137-million-compensate>

(Ferlin, Blix, and Brattström 2021) Ferlin, Maria, Johan Blix, and Emma Brattström. March 2021. "Sustainability Reporting – Need for Greater Standardisation and Transparency," March, 25.

Riksbank staff paper about the need for standardized sustainability reporting in the Swedish private sector.

<https://ypfs.som.yale.edu/library/document/sustainability-reporting-need-greater-standardisation-and-transparency>

(FI 2020a) Finansinspektionen (FI). March 13, 2020. "FI Lowers the Countercyclical Capital Buffer to Zero."

Sweden's financial authority changes the countercyclical capital buffer in response to the COVID-19 pandemic.

<https://ypfs.som.yale.edu/library/document/fi-lowers-countercyclical-capital-buffer-zero>

(FI 2020b) Finansinspektionen (FI). June 4, 2020. "Stability in the Financial System (2020:1)."

Report by Sweden's financial stability authority on the state of the financial system.

<https://ypfs.som.yale.edu/library/document/stability-financial-system-20201>

(Ingves 2021) Ingves, Stefan. May 31, 2021. "New Financial Environment - How Is the Riksbank Meeting the New Challenges?"

Speech by Governor Stefan Ingves to the Swedish Economics Association on the Riksbank's developing response to the COVID-19 pandemic.

<https://ypfs.som.yale.edu/library/document/new-financial-environment-how-riksbank-meeting-new-challenges>

(Ingves and Meyersson 2021) Ingves, Stefan, and Pernilla Meyersson. June 8, 2021. "Measures for a More Effective Scrutiny of the Riksbank."

Commentary by Governor Stefan Ingves and Deputy Head of the General Secretariat Pernilla Meyersson on proposals for the Riksdag's oversight of the Riksbank.

<https://ypfs.som.yale.edu/library/document/measures-more-effective-scrutiny-riksbank>

(Riksbank 2020b) Sveriges Riksbank (Riksbank). 2020. "The Riksbank's Business Survey- In the Wake of the Coronavirus Pandemic: 'Everything Has Changed in Two Weeks.'"

Economic Commentary 14.

Survey of businesses conducted by the Riksbank detailing COVID-19 related funding problems.

<https://ypfs.som.yale.edu/library/document/riksbanks-business-survey-wake-coronavirus-pandemic-everything-has-changed-two>

(Riksbank 2020c) Sveriges Riksbank (Riksbank). 2020. "The Riksbank's Business Survey- 'The Whole of Society Is on Hold' Companies Struggling in the Wake of the Pandemic."

Economic Commentary 13.

Survey of businesses conducted by the Riksbank detailing COVID-19 related funding problems.

<https://ypfs.som.yale.edu/library/document/riksbanks-business-survey-whole-society-hold-companies-struggling-wake-pandemic>

(Riksbank 2020k) Sveriges Riksbank (Riksbank). June 11, 2020. "How the Riksbank Creates Money in Times of Crisis."

Note on the Riksbank's procedure for creating money.

<https://ypfs.som.yale.edu/library/document/how-riksbank-creates-money-times-crisis>

(Riksbank 2022) Sveriges Riksbank (Riksbank). February 10, 2022. "The Riksbank's Extended Purchases of Securities."

Announcement of the Riksbank's asset purchase program extension.

<https://ypfs.som.yale.edu/library/document/riksbanks-purchases-securities>

(Riksgälden 2020) Riksgälden. March 27, 2020. "The Debt Office Prepares Guarantee Programme for Swedish Companies – Riksgalden.Se."

Press release announcing the Swedish National Debt Office's support scheme for businesses.

<https://ypfs.som.yale.edu/library/document/swedish-debt-office-prepares-guarantee-programme-swedish-companies>

(Skatteverket, n.d.a) Skatteverket. No date. "Reorientation Support for August 2020 to April 2021." Text. Accessed October 14, 2021.

Information on state support for companies whose cash flows were affected by the COVID-19 pandemic.

<https://ypfs.som.yale.edu/library/document/skatteverket-reorientation-support-august-2020-april-2021>

(Skatteverket, n.d.b) Skatteverket. No date. "Reorientation Support for Companies with Considerable Declines in Turnover." Accessed October 14, 2021.

Information on state support for companies whose cash flows were affected by the COVID-19 pandemic.

<https://ypfs.som.yale.edu/library/document/skatteverket-reorientation-support-companies-considerable-declines-turnover>

Reports/Assessments

(Gustafsson and von Brömsen 2021) Gustafsson, Peter, and Tommy von Brömsen. 2021. "Coronavirus Pandemic: The Riksbank's Measures and Financial Developments during Spring and Summer 2020." *Economic Commentary*.

Summary of the Riksbank's response to the COVID-19 pandemic.

<https://ypfs.som.yale.edu/library/document/coronavirus-pandemic-riksbanks-measures-and-financial-developments-during-spring>

(Hansson and Birging 2021) Hansson, Denise, and Joel Birging. June 21, 2021. "The Riksbank's Asset Purchases during the Coronavirus Pandemic."

Report from the Riksbank summarizing its asset purchases.

<https://ypfs.som.yale.edu/library/document/riksbanks-asset-purchases-during-coronavirus-pandemic>

(Nordic Trustee 2020) Nordic Trustee. 2020. "Corporate Bond Market Report."
Nordic Trustee report on the Swedish corporate bond market.
<https://ypfs.som.yale.edu/library/document/corporate-bond-market-report-2020>

(Riksbank 2020a) Sveriges Riksbank (Riksbank). 2020. "Annual Report for Sveriges Riksbank 2020."
The Riksbank's 2020 annual report, including high-level commentary on its asset purchase program.
<https://ypfs.som.yale.edu/library/document/annual-report-sveriges-riksbank-2020>

(Riksbank 2020h) Sveriges Riksbank (Riksbank). May 2020. "Financial Stability Report 2020.1."
The first of the Riksbank's biannual Financial Stability Report for 2020.
<https://ypfs.som.yale.edu/library/document/financial-stability-report-20201>

(Riksbank 2020m) Sveriges Riksbank (Riksbank). July 2020. "Monetary Policy Report July 2020."
The Riksbank's second monetary policy report following the launch of its asset purchase program in response to COVID-19.
<https://ypfs.som.yale.edu/library/document/monetary-policy-report-july-2020>

(Riksbank 2020p) Sveriges Riksbank (Riksbank). September 2020. "Monetary Policy Report September 2020."
The Riksbank's September 2020 report on monetary policy actions and their impact.
<https://ypfs.som.yale.edu/library/document/monetary-policy-report-september-2020>

(Riksbank 2020u) Sveriges Riksbank (Riksbank). November 2020. "Monetary Policy Report November 2020."
The Riksbank's November 2020 report on monetary policy, including the first reporting on its corporate bond purchases, which began in September 2020.
<https://ypfs.som.yale.edu/library/document/monetary-policy-report-november-2020>

(Riksbank 2021a) Sveriges Riksbank (Riksbank). February 2021. "Monetary Policy Report February 2021."
The Riksbank's February 2021 report on monetary policy, including updates on its asset purchases.
<https://ypfs.som.yale.edu/library/document/monetary-policy-report-february-2021>

(Riksbank 2021e) Sveriges Riksbank (Riksbank). April 26, 2021. "The Riksbank's Asset Purchases during the Third Quarter of 2021."
Report from the Riksbank summarizing its asset purchases.
<https://ypfs.som.yale.edu/library/document/riksbanks-asset-purchases-during-third-quarter-2021>

(Riksbank 2021f) Sveriges Riksbank (Riksbank). June 30, 2021. "The Riksbank's Asset Purchases during the Fourth Quarter of 2021."
Report from the Riksbank summarizing its asset purchases.

<https://ypfs.som.yale.edu/library/document/riksbanks-asset-purchases-during-fourth-quarter-2021>

(Riksbank 2021g) Sveriges Riksbank (Riksbank). September 2021. "Monetary Policy Report September 2021."

<https://ypfs.som.yale.edu/library/document/monetary-policy-report-september-2021>

(Riksbank 2021h) Sveriges Riksbank (Riksbank). November 24, 2021. "The Riksbank's Asset Purchases during the First Quarter of 2022."

Report from the Riksbank summarizing its asset purchases.

<https://ypfs.som.yale.edu/library/document/riksbanks-asset-purchases-during-first-quarter-2022>

Key Academic Papers

(Eidestedt, Forsman, and Ünlü 2020) Eidestedt, Richard, David Forsman, and Emre Ünlü. September 25, 2020. "The Funding of the Major Swedish Banks and Its Effect on Household Mortgage Rates." *Economic Commentary*.

Paper analyzing Sweden's mortgage market and bank funding.

<https://ypfs.som.yale.edu/library/document/funding-major-swedish-banks-and-its-effect-household-mortgage-rates-1>

(Riksbank 2020d) Sveriges Riksbank (Riksbank). February 2020. "The Riksbank's Measures during the Global Financial Crisis 2007-2010." *Economic Commentary* 42, February.

Riksbank paper detailing its Global Financial Crisis response.

<https://ypfs.som.yale.edu/library/document/riksbanks-measures-during-global-financial-crisis-2007-2010>

(Wollert 2020) Wollert, Stephan. October 2020. "Swedish Corporate Bonds during the Coronavirus Pandemic." *Staff Memo* 22, October.

Overview of the Swedish corporate bond market.

<https://ypfs.som.yale.edu/library/document/swedish-corporate-bonds-during-coronavirus-pandemic>

Copyright 2022 © Yale University. All rights reserved. To order copies of this material or to receive permission to reprint any or all of this document, please contact the Yale Program on Financial Stability at ypfs@yale.edu.
